

VoIP – [Write up]

Your close friend James recently received a suspicious phone call from someone claiming to be his bank. The caller asked for sensitive information, making James uneasy. Suspecting a potential Vishing (Voice Phishing) attack, you decide to investigate by capturing and analyzing the VoIP traffic.

Note: To listen to the VoIP call, you must access the machine via RDP.

Technical Overview of VoIP Protocols

Voice over Internet Protocol (VoIP) relies on two primary open-standard protocols working in tandem to enable real-time communication over IP networks: **SIP** (Session Initiation Protocol) and **RTP** (Real-time Transport Protocol). SIP operates as the signaling and control layer, responsible for establishing, managing, authenticating, and terminating connections between endpoints, while encapsulating metadata such as caller identities and timestamps. Once a session is initialized, RTP takes over as the media transport layer, converting voice signals into digital payloads and transmitting them rapidly across the network using UDP to ensure low-latency, real-time audio delivery.

| Start Investigation

Q1: How many RTP packets were in the traffic?

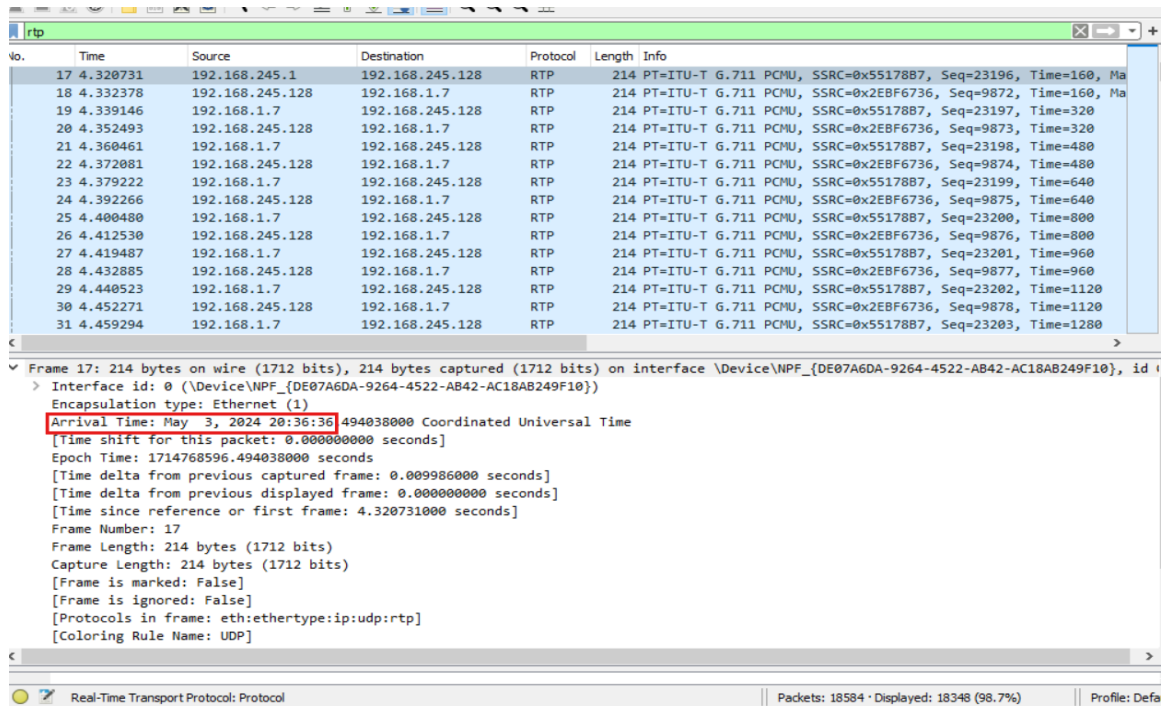
No.	Time	Source	Destination	Protocol	Length	Info
17	4.320731	192.168.245.1	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23196, Time=160, Ma
18	4.332378	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9872, Time=160, Ma
19	4.339146	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23197, Time=320, Ma
20	4.352493	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9873, Time=320, Ma
21	4.360461	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23198, Time=480, Ma
22	4.372081	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9874, Time=480, Ma
23	4.379222	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23199, Time=640, Ma
24	4.392266	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9875, Time=640, Ma
25	4.400480	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23200, Time=800, Ma
26	4.412530	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9876, Time=800, Ma
27	4.419487	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23201, Time=960, Ma
28	4.432885	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9877, Time=960, Ma
29	4.440523	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23202, Time=1120, Ma
30	4.452271	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9878, Time=1120, Ma
31	4.459294	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23203, Time=1280, Ma

Frame 19: 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits) on interface \Device\NPF_{DE07A6DA-9264-4522-AB42-AC1A8249F10}, id 0
Ethernet II, Src: VMware_ea:ic5:7c (00:50:56:ea:ic5:7c), Dst: VMware_39:70:04 (00:0c:29:39:70:04)
Internet Protocol Version 4, Src: 192.168.1.7, Dst: 192.168.245.128
User Datagram Protocol, Src Port: 4010, Dst Port: 13436
Real-Time Transport Protocol

Real-Time Transport Protocol: Protocol | Packets: 18584 · Displayed: 18348 (98.7%) | Profile: Default

For the first question, I searched for "rtp" and the number of packets appeared.

Q2: When did the fake call with James start?



No.	Time	Source	Destination	Protocol	Length	Info
17	4.320731	192.168.245.128	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23196, Time=160, Ma
18	4.332378	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9872, Time=160, Ma
19	4.339146	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23197, Time=320
20	4.352493	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9873, Time=320
21	4.360461	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23198, Time=480
22	4.372081	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9874, Time=480
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25	4.400480	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23200, Time=800
26	4.412530	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9876, Time=800
27	4.419487	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23201, Time=960
28	4.432885	192.168.245.128	192.168.1.7	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x2EBF6736, Seq=9877, Time=960
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31	4.459294	192.168.1.7	192.168.245.128	RTP	214	PT=ITU-T G.711 PCMU, SSRC=0x5517887, Seq=23203, Time=1280

Frame 17: 214 bytes on wire (1712 bits), 214 bytes captured (1712 bits) on interface \Device\NPF_{DE07A6DA-9264-4522-AB42-AC18AB249F10}, id 1
> Interface id: 0 (\Device\NPF_{DE07A6DA-9264-4522-AB42-AC18AB249F10})
Encapsulation type: Ethernet (1)
Arrival Time: May 3, 2024 20:36:36.494038000 Coordinated Universal Time
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1714768596.494038000 seconds
[Time delta from previous captured frame: 0.009986000 seconds]
[Time delta from previous displayed frame: 0.000000000 seconds]
[Time since reference or first frame: 4.320731000 seconds]
Frame Number: 17
Frame Length: 214 bytes (1712 bits)
Capture Length: 214 bytes (1712 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:rtp]
[Coloring Rule Name: UDP]

Real-Time Transport Protocol: Protocol | Packets: 18584 · Displayed: 18348 (98.7%) | Profile: Defa

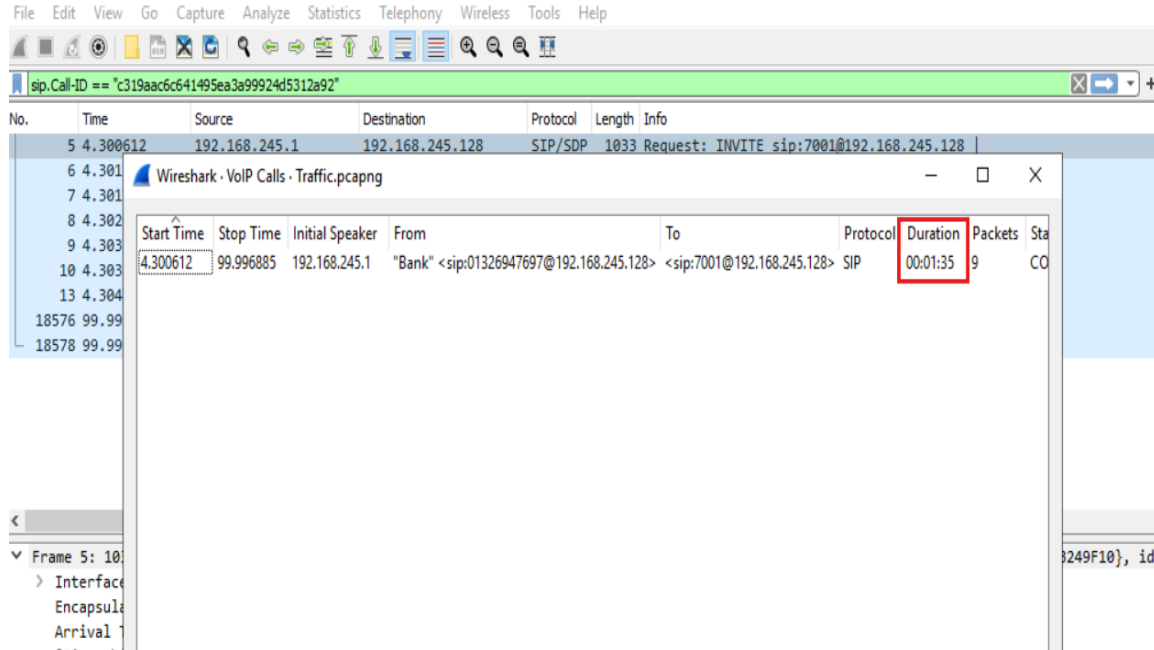
Based on our search in the first question, I checked the first displayed packet and identified the arrival time.

Q3: What is James's phone number?

I used the filter sip.Method == "INVITE" to find the James's phone number.

```
INVITE sip:7001@192.168.245.128 SIP/2.0
Via: SIP/2.0/UDP
192.168.245.1:59833;rport;branch=z9hG4bKPj002faa89bd7847bbbee1d0eb611190ed
Max-Forwards: 70
From: "Bank" <sip:01326947697@192.168.245.128>;tag=5b0ac313abce40bd82126365744463a4
To: <sip:7001@192.168.245.128>
Contact: "Bank" <sip:01326947697@192.168.245.1:59833;ob>
Call-ID: c319aac6c641495ea3a99924d5312a92
CSeq: 21391 INVITE
Allow: PRACK, INVITE, ACK, BYE, CANCEL, UPDATE, INFO, SUBSCRIBE, NOTIFY, REFER, MESSAGE,
OPTIONS
Supported: replaces, 100rel, timer, norefersub
Session-Expires: 1800
Min-SE: 90
User-Agent: MicroSIP/3.21.3
Content-Type: application/sdp
Content-Length: 335
```

Q4: How long was the call with the bank?



For the fourth question, I used the "VoIP Calls" tool under the Telephony menu to automatically display the call details and read the exact duration from the "Duration" column.

Q5: What is the phone number of the bank that James received a call from?

```
Contact: "Bank" <sip:01326947697@192.168.245.1:59833;ob>
Call-ID: c319aac6c641495ea3a99924d5312a92
CSeq: 21391 INVITE
Allow: PRACK, INVITE, ACK, BYE, CANCEL, UPDATE, INFO, SUBSCRIBE, NOTIFY, REFER, MESSAGE,
OPTIONS
Supported: replaces, 100rel, timer, norefersub
Session-Expires: 1800
Min-SE: 90
User-Agent: MicroSIP/3.21.3
Content-Type: application/sdp
Content-Length: 335
```

For the fifth question, I inspected the SIP header fields, specifically looking at the "Contact" header, to identify the spoofed bank's phone number from the SIP URI.

Executive Summary

This investigation analyzed a network traffic capture to reconstruct a vishing (voice phishing) attack where an attacker spoofed their identity as "Bank" to target a user named James. By dissecting the **SIP** signaling layer and isolating **RTP** media payloads within Wireshark, the analysis successfully mapped out the entire incident without needing to listen to the audio directly. The investigation uncovered the critical details of the fraudulent encounter, showing that the attacker initiated the call from phone number **01326947697** (IP 192.168.245.1) to James's extension **7001** on **May 3, 2024, at 20:36:36 UTC**, with the entire deceptive conversation lasting exactly **00:01:35**